

Cheng-Yuan Li

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, GPA: 4.0/4.0

Atlanta, GA

Expected May 2028

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL (PostgreSQL, MySQL), Bash

Backend & Data: REST APIs, ETL/ELT Pipelines, MongoDB (NoSQL), PySpark, Databricks, Delta Lake, AWS (S3), Google Cloud (BigQuery, Cloud Scheduler)

Tools & Practices: Git, Docker, Terraform, Linux, CI/CD, JUnit, Unit Testing, Test-Driven Development, Agile

Core CS: Data Structures & Algorithms, Concurrency, Performance Optimization

WORK EXPERIENCE

Software Engineer Intern

Gordon Food Service, Inc.

May 2026 – Aug. 2026

Atlanta, GA

- Architected an automated cost-optimization engine on Google Cloud (Python, Terraform, Cloud Scheduler) that dynamically allocates BigQuery compute resources, reducing annual infrastructure spend by **\$50,000+**.
- Cut data scanned in a 7 TB weekly pipeline by **97.8%** (to 150 GB) using incremental hash-based merges, eliminating cluster timeouts.
- Built an asynchronous pipeline processing **3 TB** of job logs across 4,000+ jobs, with automated retries and error monitoring (**99.9% uptime**).

Teaching Assistant

Georgia Institute of Technology

Jan. 2025 – Present

Atlanta, GA

- CS 4400 (Database Systems): Teach **450+** students on schema design, advanced SQL, query optimization, indexing, and transaction isolation.
- CS 1331 (Intro to OOP): Lead weekly labs for **220+ undergraduates**, covering object-oriented programming (OOP), code review, and test-driven development with JUnit.

Systems Research Intern

*A*STAR Institute of High Performance Computing*

May 2025 – Aug. 2025

Singapore

- Built an automated Python benchmarking framework to measure speed and memory use across **40+ transformer model architectures**, supporting a research manuscript.
- Analyzed multi-gigabyte GPU execution traces to identify memory bottlenecks and presented optimization findings to senior researchers.

PROJECTS

SmallDB: Key-Value Database from Scratch | C++

Aug. 2025 – Present

- Built a persistent key-value database in C++ (modeled after RocksDB) that stores data durably on disk and **recovers from crashes in under 30 ms**.
- Sustained **52,874 writes/sec** and sped up reads **89x** (to 15,515 reads/sec) using an append-only log, Bloom filters, and block caching.
- Implemented multithreaded background compaction that merges files and removes stale data without stalling incoming writes.

Kalshi Prediction Market Pipeline & Lakehouse | Python, AWS S3, Databricks

Aug. 2026 – Present

- Built an asynchronous ingestion pipeline polling 175K+ active Kalshi markets, capturing **1M+ trades/day** within a 20 req/s rate limit by tracking status changes to skip unchanged markets.
- Ensured idempotent, loss-free ingestion into date-partitioned JSONL storage using atomic checkpoint markers, verified through simulated process kills mid-stream.
- Automated daily storage replication to **AWS S3** and scheduled PySpark ETL jobs on **Databricks** to deduplicate trade executions and build market calibration tables.